

ZERO DIVISORS OF THE INTEGRAL SEDENIONS AND THE COMPACT G_2 PART IV OF THE OCTONION PROGRAM RECORD

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ABSTRACT. We study the zero divisors of the integral sedenions, the Cayley–Dickson double of Coxeter’s order of integral octonions, and their interaction with the arithmetic of the compact group G_2 . Among the 57,600 sedenions whose two octonion halves are units, exactly 7,560 are zero divisors, namely those whose halves are imaginary and orthogonal; they fall into two $G_2(\mathbb{Z})$ -orbits of sizes 6,048 and 1,512, distinguished by the type (Lipschitz or Hurwitz) of the quaternion subalgebra they generate, and the 84,672 ordered pairs (x, y) with $xy = 0$ form seven free $G_2(\mathbb{Z})$ -orbits: the integral shadow of Moreno’s theorem that the real unit zero-divisor pairs form the Lie group G_2 . Over $\mathbb{R}$ the zero-divisor manifold is $G_2/\mathrm{Sp}(1)$ and its Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$ is computed in low weights. The conic of a quaternion subalgebra modulo p meets the maximal-parahoric class set of the compact G_2 at p in a pattern governed by the octahedral group S_4 : for $p \in \{3, 5, 7, 11\}$ each conic lies in a single class determined by the type of the zero divisor, and for $p \geq 13$ it splits into S_4 -orbits, with class stabilisers $192/m$ and $48/m$; this is proved for all $p \geq 5$. Zero-divisor periods of algebraic modular forms on G_2 at level 3 are defined; they are the D^1 -periods of Gan and Savin’s exceptional see-saw, and the period along the conic separates Iwahori-new forms from the old ones (through the degeneracy map), which are identified exactly as Gan–Gurevich-type lifts of the level-3 newforms of weights 6 and 10 with the classical form on D^1 . Three constructions of a sedenion spectral geometry in the sense of Farnsworth and Boyle are shown to admit no finite Dirac operator beyond a scalar mass, and the Euler product of the norm-counting zeta function is shown to survive exactly up to the sedenions along the Cayley–Dickson ladder. All computations and code are deposited in the program’s repository.

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1. INTRODUCTION AND SUMMARY

Parts I and II of this record [1, 2] concern class sets and Hecke modules of the compact groups G_2 and F_4 built from Coxeter's integral octonions $\mathbb{O}_{\mathbb{Z}}$ (the E_8 lattice). The present part goes one rung further on the Cayley–Dickson ladder, to the sedenions $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$, and asks what the first failure of division has to do with the arithmetic developed so far. The answer is more structured than expected.

- (a) The integral sedenions $\mathbb{S}_{\mathbb{Z}} = \mathbb{O}_{\mathbb{Z}} \oplus \mathbb{O}_{\mathbb{Z}}\ell$ form the lattice $E_8 \oplus E_8$. Their zero divisors with unit halves are *exactly* the elements $a + b\ell$ with a, b imaginary orthogonal units (Theorem 2.1); there are 7,560 of them, each with a 4-dimensional annihilator, and they form two $G_2(\mathbb{Z})$ -orbits separated by the type of the quaternion subalgebra $\langle 1, a, b, ab \rangle$. The ordered zero-divisor pairs form seven free $G_2(\mathbb{Z})$ -orbits (Theorem 2.2).
- (b) Over $\mathbb{R}$ the unit zero divisors form $G_2/\mathrm{Sp}(1)$ and Moreno's G_2 is the sphere bundle of their annihilators. The Hecke algebra $\mathcal{H}(G_2, \mathrm{Sp}(1))$, a non-commutative algebra with continuous families of idempotents, has blocks of sizes 1, 3, 3, 6, 8, 10, 6 on the representations of dimensions 1, 7, 14, 27, 64, 77, 77' (Section 3).
- (c) The conic law (Theorem 4.4): the isotropic lines of $\mathrm{Im} \mathbb{H}_z \otimes \mathbb{F}_p$ form a conic $\cong \mathbb{P}^1(\mathbb{F}_p)$, and its intersection with the maximal-parahoric class set of the compact G_2 at p is governed by the octahedral group S_4 . The two zero-divisor types index two classes for $p = 3, 5, 7, 11$ and the conics split into cube-orbits for $p \geq 13$.
- (d) Zero-divisor periods of algebraic modular forms (Section 5) are the D^1 -periods of the Gan–Savin see-saw [6]; the period along the conic vanishes exactly on the Iwahori-new forms of weight $2\omega_1$ at level 3, while their periods on other orbits do not.
- (e) Negatives (Section 6): in the Farnsworth–Boyle framework of non-associative spectral geometry [8, 9] the sedenions admit, in all constructions tried, no finite Dirac operator beyond a scalar mass; the sedenion analogue of the Albert algebra is not Jordan; and the Euler product of the norm-counting zeta function dies one doubling past the sedenions, where Ramanujan's Δ first enters.

Everything in (a)–(d) was computed exactly (integer arithmetic, finite groups, or linear algebra at machine precision with independent checks); the conic law is proved for all $p \geq 5$ by a structural reduction to a finite computation (Section 4).

2. THE INTEGRAL SEDENIONS AND THEIR ZERO DIVISORS

Let $\mathbb{O}_{\mathbb{Z}}$ be Coxeter's maximal order of integral octonions, with norm N , trace tr and the 240 units. The Cayley–Dickson double $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ has product $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$, conjugation $\overline{(a, b)} = (\bar{a}, -b)$ and norm $N(a, b) = N(a) + N(b)$; we take $\mathbb{S}_{\mathbb{Z}} = \mathbb{O}_{\mathbb{Z}} \oplus \mathbb{O}_{\mathbb{Z}}\ell$. On 200 random triples the laws behave as classically known: $\mathbb{S}$ is flexible and power-associative; it is not commutative, associative or alternative, and $N(xy)/(N(x)N(y))$ ranged over $[0.709, 1.316]$.

Theorem 2.1. *Let $x = (a, b) \in \mathbb{S}_{\mathbb{Z}}$ with $N(a) = N(b) = 1$. Then x is a zero divisor if and only if a and b are both imaginary ($\mathrm{tr} = 0$) and orthogonal. There are 7,560 such elements; each has a 4-dimensional annihilator $\{y : xy = 0\}$. No unit of $\mathbb{S}_{\mathbb{Z}}$ (norm 1) is a zero divisor.*

Proof. Exhaustive computation over all $240^2 = 57,600$ pairs (a, b) of units: the left-multiplication matrix L_x is singular exactly for the 7,560 pairs with $\mathrm{tr} a = \mathrm{tr} b = 0$ and $\langle a, b \rangle = 0$ (the other

seven combinations of the three conditions never give a zero divisor), and its kernel has dimension 4 in every case. A unit has one half zero and all 230,400 products of units have norm 1. $\square$

The real version (imaginary, orthogonal, equal norm) is due to Moreno and to Khalil–Yiu; the count and the dimension 4 reproduce Moreno’s dimension count $11 + 3 = 14 = \dim G_2$ for the set of unit zero-divisor pairs [3].

Theorem 2.2. *The group $G_2(\mathbb{Z}) = \text{Aut}(\mathbb{O}_{\mathbb{Z}})$ (order 12,096) acts on the 7,560 zero divisors with two orbits, of sizes 6,048 and 1,512, with stabilisers of orders 2 and 8. The quaternion subalgebra $\mathbb{H}_z = \langle 1, a, b, ab \rangle \cap \mathbb{O}_{\mathbb{Z}}$ contains 8 units (Lipschitz type) on the large orbit and 24 (Hurwitz type) on the small one, and these are exactly the numbers of zero divisors y in the annihilator of x among the 7,560. The $84,672 = 7 \cdot 12,096$ ordered pairs (x, y) of these zero divisors with $xy = 0$ form seven free $G_2(\mathbb{Z})$ -orbits.*

Proof. Computation with the 12,096 automorphisms in the Coxeter basis (each verified to be a sedenion automorphism acting diagonally on the two halves); the orbit and stabiliser counts are exact. The map $(a, b) \mapsto (a, -b)$ is an automorphism of $\mathbb{S}$; the half-swap preserves the zero-divisor set but is not an automorphism. $\square$

Remark 2.3. Moreno’s theorem says the real unit zero-divisor pairs form a space homeomorphic to G_2 ; Theorem 2.2 is its integral shadow: seven copies of $G_2(\mathbb{Z})$, acted on freely.

3. THE ZERO-DIVISOR MANIFOLD AND ITS HECKE ALGEBRA

Over $\mathbb{R}$, a unit zero divisor is a pair (a, b) of orthonormal imaginary octonions. G_2 acts transitively on such pairs, so the zero-divisor manifold is $ZD = G_2/K$ with K the stabiliser of one pair.

Proposition 3.1. *K is 3-dimensional with Lie algebra $\mathfrak{su}(2)$; it fixes $\mathbb{H}_z$ pointwise and acts on the 4-dimensional annihilator of $z = (a, b)$ as $\text{Sp}(1)$ acts on $\mathbb{H}$ (the commutant of K on the annihilator is 4-dimensional). Hence $ZD = G_2/\text{Sp}(1)$ is the Stiefel manifold of orthonormal pairs in $\mathbb{R}^7$, of dimension 11, and Moreno’s G_2 is the unit-sphere bundle of the annihilators over it.*

Proof. Computed from the 14 derivations of $\mathbb{O}$ in an orthonormal basis: the derivations killing a and b form a 3-dimensional subalgebra, which kills $\langle 1, a, b, ab \rangle$ and preserves the annihilator, on which its commutant has dimension 4. $\square$

In Gan–Savin’s notation [6], $K = D^1$ is the group of norm-one elements of the quaternion subalgebra $D = \mathbb{H}_z$, acting by $g \cdot (a, b) = (a, b\bar{g})$; it is the pointwise stabiliser of D in $\text{Aut}(\mathbb{O})$.

The Hecke algebra of the homogeneous space is $\mathcal{H}(G_2, K) = \bigoplus_{\pi} M_{m_{\pi}}(\mathbb{C})$ with $m_{\pi} = \dim \pi^K$ (Frobenius reciprocity: m_{π} is the multiplicity of π in $L^2(ZD)$). Decomposing $7 \otimes 7$, $7 \otimes 14$, $\text{Sym}^3 7$ and $\text{Sym}^2 14$ by the Casimir and counting K -fixed vectors:

π	1	7	14	27	64	77	77'
$m_{\pi} = \dim \pi^K$	1	3	3	6	8	10	6

The 7 gives 3 because K fixes exactly $\text{Im } \mathbb{H}_z \subset \text{Im } \mathbb{O}$; the 14 gives 3 because the centraliser of K in $\mathfrak{g}_2$ is the other $\mathfrak{su}(2)$ of $\mathfrak{so}(4) \subset \mathfrak{g}_2$. Since (G_2, K) is not a Gelfand pair, the idempotents of each block form continuous families (Grassmannians). The representation theory here is classical; the identification of ZD with the sedenion zero divisors is what makes it ours. Every weight in the Path A table of compact G_2 forms (Part II) has $m_{\pi} > 0$, which is what makes the periods of Section 5 possible.

4. QUATERNION SUBALGEBRAS AND THE CONIC LAW

4.1. Quaternion subalgebras of the Coxeter order. Every zero divisor $z = (a, b)$ determines $\mathbb{H}_z = \langle 1, a, b, ab \rangle \cap \mathbb{O}_{\mathbb{Z}}$, a quaternion order of the definite quaternion algebra it spans. Counting ordered orthogonal imaginary unit pairs per subalgebra ($6 \cdot 4 = 24$ in both cases) gives:

Proposition 4.1. $\mathbb{O}_{\mathbb{Z}}$ contains 63 quaternion subalgebras of Hurwitz type (24 units) and 252 of Lipschitz type (8 units). Their stabilisers in $G_2(\mathbb{Z})$ have orders 192 and 48; the pointwise stabilisers have orders 8 and 2 (the stabilisers of the zero divisors of Theorem 2.2), and in both cases the quotient acting faithfully on $\text{Im } \mathbb{H}_z$ is the octahedral group S_4 .

Proof. $1512/24 = 63$, $6048/24 = 252$; $12096/63 = 192$, $12096/252 = 48$; the pointwise stabilisers were computed directly, and $192/8 = 48/2 = 24 = |S_4|$, with S_4 the full rotation group of the cube $\{\pm i, \pm j, \pm k\}$, realised in both cases by the computation of Section 5 (orbit sizes 6, 8, 12 below). $\square$

Proposition 4.2. The imaginary integral units are the 126 roots of the E_7 lattice $\text{Im } \mathbb{O}_{\mathbb{Z}}$, and the 7,560 zero divisors of Theorem 2.1 are exactly the ordered pairs of orthogonal E_7 roots (60 roots are orthogonal to any given root, a D_6). The zero-divisor mass $7560/12096 = 5/8$ is thus a Siegel representation number of the binary form $2I_2$ by E_7 divided by $|G_2(\mathbb{Z})|$, and the zero divisors of each norm are governed by Siegel's local-global formula for representations by E_7 , even though their counting series has no Euler product of its own.

Proposition 4.3. For a zero-divisor pair $x = (a, b)$, $y = (c, d)$ with $xy = 0$ one has $d = \pm(ac)b$, so the pair is the triple (a, b, c) with c a unit perpendicular to $\mathbb{H}_x$. The seven free $G_2(\mathbb{Z})$ -orbits of Theorem 2.2 are separated by the types (Hurwitz 24 or Lipschitz 8) of the five quaternion subalgebras $\langle a, b \rangle, \langle c, d \rangle, \langle a, c \rangle, \langle b, c \rangle, \langle ab, c \rangle$: the seven realised type-patterns are $(8, 8, 8, 24, 8)$, $(8, 8, 24, 8, 8)$, $(8, 24, 8, 8, 24)$, $(8, 24, 24, 24, 24)$ for Lipschitz x and $(24, 8, 8, 24, 8)$, $(24, 8, 24, 8, 8)$, $(24, 24, 8, 8, 24)$ for Hurwitz x , with equal counts within each type as the freeness requires.

4.2. The conic. Fix an odd prime p . The space $\text{Im } \mathbb{O} \otimes \mathbb{F}_p$ is a nondegenerate 7-dimensional quadratic space; its isotropic lines number $(p^6 - 1)/(p - 1)$ and form the maximal-parahoric level- p structures of the compact G_2 , so that the class set is $X_p = G_2(\mathbb{Z}) \backslash \{\text{isotropic lines}\}$ (Part I). For a zero divisor z , $\text{Im } \mathbb{H}_z \otimes \mathbb{F}_p$ is a nondegenerate ternary space, so its isotropic lines form a conic $C_z \cong \mathbb{P}^1(\mathbb{F}_p)$ with $p + 1$ points, on which $\text{Stab}(\mathbb{H}_z)$ acts through $S_4 \subset \text{PGL}_2(\mathbb{F}_p)$.

Theorem 4.4 (Conic law). *Let $p \geq 5$ and z a zero divisor. The $G_2(\mathbb{Z})$ -classes of the points of C_z are exactly the S_4 -orbits on $\mathbb{P}^1(\mathbb{F}_p)$, whose sizes lie in $\{6, 8, 12, 24\}$ (cube faces, vertices, edges, and generic), and the stabiliser in $G_2(\mathbb{Z})$ of a conic point in an orbit of size m has order $192/m$ (Hurwitz type) or $48/m$ (Lipschitz type). The number of classes a conic meets is the number of S_4 -orbits on $\mathbb{P}^1(\mathbb{F}_p)$, which by Burnside's lemma is*

$$\frac{1}{24} \left((p+1) + 3f_{-1} + 6f_{-2} + 8f_3 + 6f_4 \right), \quad f_{-1} = 2 \text{ if } \left(\frac{-1}{p}\right) = 1, \quad f_{-2} = 2 \text{ if } \left(\frac{-2}{p}\right) = 1, \quad f_3 = 2 \text{ if } p \equiv 1 \pmod{3}, \quad f_4 = 2 \text{ if } p \equiv 1 \pmod{4},$$

and 0 otherwise (the face, edge and vertex rotations of the cube have 2 or 0 fixed points on the conic according to these conditions). Equivalently, the conic carries a 6-point orbit (stabilisers 8 and 32) exactly when $p \equiv 1 \pmod{4}$, an 8-point orbit (6 and 24) exactly when $p \equiv 1 \pmod{3}$, a 12-point orbit (4 and 16) exactly when $\left(\frac{-2}{p}\right) = 1$, and all remaining points lie in 24-point orbits (2 and 8); this was verified directly for all $p \leq 41$. The orbit structure is thus Eisenstein data of level 24: the number of classes a conic meets is $\frac{p+1}{24} + \frac{3}{4}[p \equiv 1 \pmod{4}] + \frac{1}{2}[\left(\frac{-2}{p}\right) = 1] + \frac{2}{3}[p \equiv 1 \pmod{3}]$, and its Dirichlet series over p is a combination of $\zeta(s-1)$, $\zeta(s)$ and the L -functions of the characters of conductors 3, 4 and 8, while the cuspidal content of the zero divisors lies entirely in the periods

of Section 5. In particular each conic lies in a single class exactly for $p \in \{5, 7, 11, 23\}$ ($p = 23$ is the regular orbit, 24 points with stabilisers 2 and 8), and for $p \geq 29$ never; for $p = 3$ the conic (4 points) also lies in a single class, with the exceptional stabilisers 432 and 36. The count depends only on p modulo 24.

Lemma 4.5. *For $p \geq 5$, an isotropic line lying on the conic of a quaternion subalgebra of given type lies on no other conic of that type.*

Proof. Let $\mathbb{H}_1 \neq \mathbb{H}_2$ be suborders of the same type, $M_i = \text{Im } \mathbb{H}_i$ (rank 3). Since $\mathbb{H}_i = B_i \cap \mathbb{O}_{\mathbb{Z}}$ for the $\mathbb{Q}$ -subalgebra B_i it spans, $B_1 \neq B_2$, and two distinct quaternion subalgebras of $\mathbb{O}$ meet in a subalgebra of dimension ≤ 2 (a 3-dimensional subalgebra of a quaternion algebra does not exist); hence $M_1 \cap M_2$ has rank ≤ 1 . The conics C_1, C_2 modulo p share a point only if $\bar{M}_1 \cap \bar{M}_2 \subset \text{Im } \mathbb{O} \otimes \mathbb{F}_p$ contains a nonzero isotropic vector. Now $\dim_{\mathbb{F}_p}(\bar{M}_1 \cap \bar{M}_2) = 6 - \text{rank}_p(M_1 + M_2)$, and $\text{rank}_p(M_1 + M_2) = \text{rank}(M_1 + M_2)$ unless p divides the index of $M_1 + M_2$ in its saturation. So either (i) p divides that index, or (ii) $\bar{M}_1 \cap \bar{M}_2$ is the reduction of $M_1 \cap M_2$, which is 0 or $\mathbb{Z}v$ with v a common imaginary vector, isotropic mod p only if $p \mid N(v)$.

Computation over all $252 \cdot 251/2 = 31,626$ Lipschitz pairs and $63 \cdot 62/2 = 1,953$ Hurwitz pairs (code `g2table/lemma43.py`): the saturation indices take the values 1, 2, 3, 4, 5, 6, 8, 10 (Lipschitz) and 1, 2, 3, 4, 5, 6, 7, 8 (Hurwitz), so their prime factors lie in $\{2, 3, 5, 7\}$; the pairs with $\text{rank}(M_1 + M_2) = 5$ (6,426 Lipschitz, 1,197 Hurwitz) have common directions of norm 1 or 2 (Lipschitz) and 1 or 3 (Hurwitz). Hence for $p \geq 11$ case (i) never occurs and case (ii) gives no isotropic vector: the conics are disjoint. For $p = 5$ and $p = 7$, case (i) occurs for 15 + 75 pairs at $p = 5$ and 23 Hurwitz pairs at $p = 7$; for each of these the intersection $\bar{M}_1 \cap \bar{M}_2$ was computed over $\mathbb{F}_p$ and found to contain no isotropic vector, while case (ii) is excluded as before since $p \nmid N(v) \in \{1, 2, 3\}$. This proves the lemma for all $p \geq 5$. (For $p = 3$ it fails: $3 \mid N(v)$ for the Hurwitz pairs with a common direction of norm 3, and the conics do share points, which is why the stabilisers 432, 36 at $p = 3$ exceed $192/4$ and $48/4$.) $\square$

Proof of Theorem 4.4 from Lemma 4.5. Let $\ell \in C_z$ and $g \in G_2(\mathbb{Z})$ with $g\ell \in C_z$. Then $g\ell$ lies on the conics of $\mathbb{H}_z$ and of $g\mathbb{H}_z$, both of the type of z ; by the lemma $g\mathbb{H}_z = \mathbb{H}_z$, so $g \in \text{Stab}(\mathbb{H}_z)$. Hence the $G_2(\mathbb{Z})$ -orbit of ℓ meets C_z exactly in its $\text{Stab}(\mathbb{H}_z)$ -orbit, which is its S_4 -orbit since the pointwise stabiliser acts trivially on C_z , and the stabiliser of ℓ in $G_2(\mathbb{Z})$ equals its stabiliser in $\text{Stab}(\mathbb{H}_z)$, of order $|\text{Stab}(\mathbb{H}_z)|/m$. The orbit sizes of $S_4 \subset \text{PGL}_2(\mathbb{F}_p)$ on $\mathbb{P}^1(\mathbb{F}_p)$ are $24/|\text{point stabiliser}|$ with point stabilisers the cyclic subgroups of orders 4, 3, 2, 1, giving 6, 8, 12, 24; S_4 is transitive on $\mathbb{P}^1(\mathbb{F}_p)$ iff $p+1 \in \{6, 8, 12\}$ (the case 4 being $p = 3$, where the lemma fails and the stabilisers are larger). $\square$

Computation 4.6. Independently of the proof, exact orbit computations for $p = 3, 5, 7, 11, 13, 17, 19$ (four to forty zero divisors of each type per prime) give the class structure of the conic points shown in Table 1; the stabiliser identity $|\text{Stab}(\ell)| = |\text{Stab}(\mathbb{H}_z)|/m$ holds in every case with $p \geq 5$, as the theorem predicts. The classes touched were also matched against the full class sets X_5 (3 classes, stabilisers 32, 8, 6: the 6-class is touched by no zero divisor) and X_7 (8 classes).

Remark 4.7. The incidence counts are now explained: $63(p+1)$ Hurwitz and $252(p+1)$ Lipschitz incidences fall on classes of sizes $12096m/192$ and $12096m/48$, one conic through each point.

5. ZERO-DIVISOR PERIODS

5.1. Definition. Let f be an algebraic modular form of weight π for the compact G_2 at a parahoric level at $p = 3$: a function on the level-3 structures x (isotropic lines, planes or flags

p	$p + 1$	Lipschitz: (points, stabiliser)	Hurwitz: (points, stabiliser)
3	4	(4, 36)	(4, 432)
5	6	(6, 8)	(6, 32)
7	8	(8, 6)	(8, 24)
11	12	(12, 4)	(12, 16)
13	14	(8, 6) + (6, 8)	(8, 24) + (6, 32)
17	18	(12, 4) + (6, 8)	(12, 16) + (6, 32)
19	20	(12, 4) + (8, 6)	(12, 16) + (8, 24)

TABLE 1. Class structure of the conic points of a zero divisor; stabilisers are $48/m$ and $192/m$ for $p \geq 5$.

mod 3) with $f(x\gamma) = \pi(\gamma)^{-1}f(x)$, in the conventions of [2]. For a zero divisor z with continuous stabiliser K_z , and $u \in \pi^{K_z}$, put

$$P_{z,u}(f) = \sum_{x \text{ compatible with } z} \langle f(x), u \rangle, \quad x \text{ compatible} \iff \text{the line of } x \text{ lies in } \text{Im } \mathbb{H}_z \text{ mod } 3,$$

and more generally, for any orbit O of the finite group $\text{Stab}(\mathbb{H}_z)$ on the structures, $P_{z,O}(f) = \text{proj}_{\pi^{K_z}} \sum_{x \in O} f(x)$. Every zero divisor has exactly 4 compatible lines (the conic at $p = 3$).

5.2. Trivial weight. At the maximal parahoric level X_3 has two classes (stabilisers 432, 36) and T_2 -eigenvalues 126 (Eisenstein) and 9 (the lift of 3.6.a.a). By Theorem 4.4 the four conic points of a Hurwitz zero divisor lie in the 432-class and those of a Lipschitz one in the 36-class, so $P_z(f) = 4f(\text{class of } z)$: for the cusp form, normalised $f = (-1, 1/12)$, the periods are -4 and $1/3$.

5.3. Weights 14 and 27. Weight 14 (adjoint): one form at level 3 ($T_2 = -18$, at the plane and flag levels), whose conic periods vanish for both types. Weight 27 ($2\omega_1$): at the Iwahori level a 7-dimensional space with $T_2 \in \{40.5 (\times 2), 0 (\times 3), -10.768, 6.268\}$, the irrational pair being the roots of $x^2 + 4.5x - 67.5$ and new at the Iwahori level; the others descend to the maximal parahoric levels, where $(T_2, T_5) = (81/2, 2034/5)$ and $(0, 11466/25)$ (twice), computed with the calibration (9, 810) and (126, 19530) in trivial weight.

Computation 5.1. (i) The conic period $P_{z,u}$ vanishes identically on the Iwahori-new forms and is nonzero on all old forms, for both types. (ii) Over all $\text{Stab}(\mathbb{H}_z)$ -orbits (51 orbits for Lipschitz, 21 for Hurwitz type), the Iwahori-new forms have nonzero $P_{z,O}$ on 41/51 resp. 16/21 orbits, vanishing exactly on the orbits containing the conic flags (and a few others); the old forms are nonzero on every orbit. (iii) For the Iwahori-new forms the sum of f over the four flags above *each* conic point vanishes separately, before any projection.

Remark 5.2 (what the conic vanishing is). Item (iii) explains item (i) completely: the sum of f over the planes containing a fixed line is the degeneracy map from the Iwahori level to the maximal-parahoric line level, and Iwahori-new forms lie in its kernel for *every* line. The conic period factors through this degeneracy map, so its vanishing on new forms is the definition of newness, not a property of the zero divisors; the Hurwitz/Lipschitz type and the K -projection play no role in it. The content of zero-divisor periods is therefore (a) at the maximal parahoric level, where the conic law (Theorem 4.4) governs them, and (b) in orbit-periods $P_{z,O}$ that do not factor through degeneracy maps, which is where the new forms are nonzero, in accordance with the global expectation of the next subsection.

5.4. The maximal parahoric level 7 and the selection rule. At the maximal parahoric level $p = 7$ the class set has 8 classes (stabilisers 2, 18, 42, 6, 6, 2, 6, 24), and the conics of the Hurwitz and Lipschitz zero divisors lie in the 24-class and in one 6-class (Theorem 4.4). Since the $p + 1$ conic points form a single orbit of the finite group $\text{Stab}(\mathbb{H}_z)$ when $p \leq 11$, the conic sum $\sum_{\ell \in C_z} f(\ell)$ is the projection of $f(\ell_0)$ onto the $\text{Stab}(\mathbb{H}_z)$ -fixed vectors of V_π , which gives a selection rule: $\dim V_\pi^{\text{Stab}(\mathbb{H}_z)}$ is 1 for the trivial weight, 0 for the 7 and the 14, and 1 (Hurwitz) or 2 (Lipschitz) for the 27. The rule for all weights up to 77, computed from the characters of $\text{Stab}(\mathbb{H}_z)$ on the eigenvalues of its elements on the 7, is

weight	1	7	14	27	64	77	77'
$\dim V_\pi^{\text{Stab}(\mathbb{H}_z)}$, Lipschitz ($ \text{Stab} = 48$)	1	0	0	2	1	1	3
$\dim V_\pi^{\text{Stab}(\mathbb{H}_z)}$, Hurwitz ($ \text{Stab} = 192$)	1	0	0	1	0	0	2

so from weight 64 on the two types differ: in weights 64 and 77 every form is dark on Hurwitz conics and (generically) lit on Lipschitz ones. This was verified in weight 64 at level 7 (97 forms, the 64 cut out of $7 \otimes 14$ by the character projector over $G_2(\mathbb{Z})$): all 97 conic sums vanish for the Hurwitz type and none for the Lipschitz type. Accordingly all conic sums vanish identically in weights 7 and 14 for every form, and in weight 27 (50 forms at this level) the periods are nonzero for every form and both types; in trivial weight (7 tempered cusp forms) likewise. So zero-divisor periods do not vanish on tempered forms: they are generic nonzero functionals, in accordance with the nonvanishing predicted by the see-saw, and their content lies in the conic law and the selection rule rather than in any vanishing.

The same computation extends Theorem 5.3 to level 7: the five newforms of $S_{10}(\Gamma_0(7))$ appear in weight $2\omega_1$ at the maximal parahoric level with exactly the predicted (T_2, T_3) , namely (58.351, 116.103), (37.021, 188.629), (35.169, 28.970) (multiplicity 2), (14.331, 164.808) (multiplicity 2) and (1.378, 80.601); the first two violate the tempered bound, the others pass it. In trivial weight the level-7 lifts of weight-6 newforms have no fixed vector at this parahoric and live at the plane parahoric.

5.5. Interpretation. In [6] the pair $D^1 \subset \text{Aut}(\mathbb{O})$ sits in a see-saw inside the adjoint E_7 of the Albert algebra, with $\text{Aut}(\mathbb{O}) \times \text{PGSp}_6$ and $D^1 \times G_D$ (G_D of type D_6) as dual pairs; the theta lift of the trivial representation of D^1 is a residue of a Siegel Eisenstein series on G_D , and by Pollack's integral and Chenevier's identity $L(s, \pi, \text{Spin}) = \zeta(s)L(s, \pi', \text{Std})$ for G_2 -parameters, the D^1 -period of a form on the compact G_2 is governed by $L(1, \pi', \text{Std}) \neq 0$. Our periods are D^1 -periods with specific local data at 3. Computation 5.1(ii) is in accordance: the genuine (Iwahori-new) forms have nonzero D^1 -periods on orbits that do not factor through degeneracy maps; the vanishing along the conic is the degeneracy map itself (Remark above). The old forms are identified. Writing $\text{SO}(4) = (\text{SL}_2^{\text{long}} \times \text{SL}_2^{\text{short}})/\mu_2 \subset G_2$ for the stabiliser of a quaternion subalgebra, with $7 = (2, 2) \oplus (1, 3)$ and $\text{SL}_2^{\text{long}} = D^1$, consider the Arthur-type parameter $\psi_f = \varphi_f \boxtimes \nu$ with a classical newform f of weight k on the long-root SL_2 and the Arthur SL_2 on the short root. Since the Hecke eigenvalue of an algebraic modular form is the Satake transform of the unitary local representation (the weight enters only at infinity), $T_p = p^3 \chi_7(s_p) - 1$ in every weight, and here

$$\chi_7(s_p) = \left(p + 1 + \frac{1}{p}\right) + (p^{1/2} + p^{-1/2}) \frac{a_p(f)}{p^{(k-1)/2}}, \quad T_p = \frac{p^3(p^2 + p + 1)}{p} - 1 + \frac{(p + 1) a_p(f) p^3}{p^{k/2}}.$$

Theorem 5.3. *At level 3 the trivial-weight cusp form $(T_2, T_5, T_7) = (9, 810, 2472)$ is $\psi_{3.6.a.a}$ ($a_2, a_5, a_7 = -6, 6, -40$), and the weight- $2\omega_1$ old forms are $\psi_{3.10.a.a}$ ($a_2, a_5 = 18, -1530$; $T_2 = 81/2$, $T_5 = 2034/5$) and $\psi_{3.10.a.b}$ ($a_2, a_5 = -36, -1314$; $T_2 = 0$, $T_5 = 11466/25$, multiplicity 2 at*

the maximal parahoric level), with the weight rule $k = 6 + 2a$ for G_2 -weight $a\omega_1$. All six (resp. four) Hecke eigenvalues agree exactly; the prime 7 was not used in finding the dictionary.

These are the Gan–Gurevich non-tempered forms [5] seen on the compact G_2 : the classical form lives on D^1 , which is why their D^1 -periods are large, and the two weight-10 forms have opposite Atkin–Lehner signs at 3 ($a_3 = \pm 81$), a natural source of the multiplicity difference (to be confirmed against Arthur’s multiplicity formula). No level-one forms exist in weights 14, 27, 77’ ($G_2(\mathbb{Z})$ -invariants vanish). The Iwahori-new irrational pair is not of this type and is a genuine (tempered) form.

6. NEGATIVES AND BOUNDARIES

6.1. Sedenion spectral geometries. Farnsworth and Boyle [8, 9] extend Connes’ spectral triples to non-associative algebras: the Hilbert space must be a bimodule in the algebra’s own class (every identity surviving the replacement of one algebra element by a vector), the order conditions become vanishing associators in $B = \Omega A \oplus H$, and the gauge group is $\text{Aut}(A)$. For the sedenions acting on themselves: the flexible order-zero law holds and the alternative ones fail; the derivation algebra is 14-dimensional and inner (inside $\mathfrak{so}(16) \oplus \mathbb{R}$); the fermions decompose as $2(1+7)$ under G_2 with the S_3 of $\text{Aut}(\mathbb{S}) = G_2 \times S_3$ as a family symmetry; the commutant of the left action is trivial, forcing $\gamma = \pm 1$ and $D_F = 0$. On two copies $\mathbb{S} \oplus \mathbb{S}$ with $\gamma = \text{diag}(1, -1)$, the flexible order-one condition forces D_F to be a multiple of the identity for either real structure; with the opposite bimodule on the second copy, no D_F survives; on the complex sedenions $\mathbb{C} \otimes \mathbb{S}$ with real-linear D_F , only complex scalars survive. The sedenion analogue $H_3(\mathbb{S})$ of the Albert algebra violates the Jordan identity (relative size 0.96; $H_3(\mathbb{O})$ satisfies it to 10^{-15}), so the Jordan-geometry route [9] excludes the sedenions and admits the Albert algebra. In every construction the zero divisors do not enter the action.

6.2. Euler products along the ladder. The norm-counting Dirichlet series $\sum r(n)n^{-s}$ has an Euler product iff r is multiplicative. It does for $\mathbb{Z}$, $\mathbb{Z}[i]$, the Hurwitz quaternions (D_4), the Coxeter octonions (E_8 , $r = 240\sigma_3$) and the integral sedenions ($E_8 \oplus E_8$, $r = 480\sigma_7$) — the last only because $M_8(\text{SL}_2(\mathbb{Z}))$ has no cusp forms — and fails for the integral 32-ions (E_8^4), since $E_4^4 = E_{16} + \frac{3456000}{3617} E_4 \Delta$. The counting zeta function of the 32-ions, $\frac{16320}{3617} \zeta(s) \zeta(s-15) + c L(E_4 \Delta, s)$, has two zeros on its critical line $\Re s = 8$ below $t \approx 15$ and none up to $t = 100$, where the Eisenstein part dominates by 10^5 – 10^6 ; the Leech lattice’s $\zeta(s) \zeta(s-11) - L(\Delta, s)$ has no zeros at all on $\Re s = 6$ to $t = 120$, its 57 zeros to height 198 lying near $\Re s \in [11.1, 12.5]$ with spacings more rigid than GUE. The product over primes is lost exactly where Δ enters.

7. OPEN PROBLEMS

- (1) A conceptual proof of Lemma 4.5 replacing the finite computation at $p = 5, 7$ (the primes 5 and 7 divide some saturation indices without producing isotropic intersections).
- (2) The multiplicities of the Arthur-type lifts of Theorem 5.3 against Arthur’s multiplicity formula ($a_3 = \pm 81$), and the identification of the Iwahori-new tempered pair through its exceptional theta lift to genus-3 Siegel forms of level 3.
- (3) The 32-ions; the Okubo integral orders of [12] and their idempotent geometry against the conic law.
- (4) Part III: the J_E half of the F_4 class set at level 5 (88% of the mass found; exact deficits 720799/96768 and 3).

APPENDIX A. CODE AND DATA

Repository `chrisyounge-create/Cayley-Agent-Repository`: `sedenion.py` (doubling, laws), `sedenion_geometry.py`, `sedenion_step2.py`, `sedenion_step3.py`, `sedenion_step3_complex.py` (spectral geometries), `zd_hecke.py` (stabiliser, annihilator, multiplicities), `g2table/zd_periods.py`, `g2table/zd_periods_vec.py` (periods), `g2table/lemma43.py` (proof of Lemma 4.5), `g2table/zd_periods_level7.py` (level-7 periods and lifts), `beurling.py` (prime-correlation model). Dropbox notes `RH/SEDENIONS.md` through `RH/NEXT_RUN_ITEMS_1_TO_4.md` contain the session records.

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